

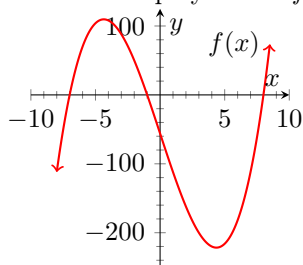
1. Evaluate the following:

a. -2^{-3}

b. $\left(-\frac{2}{3}\right)^{-2}$

c. $8^{-4/3}$

d. $f(x) = 5^{2-3x}$, find $f(1)$.

2. Consider the polynomial $f(x)$ shown in the graph.a. What are the zeros of $f(x)$?b. Does $f(x) = (x - 7)(x + 8)(x + 1)$?

- If the factor $x - c$ occurs k times in the complete factorization of a polynomial $P(x)$, then c is called a root/zero of $P(x)$ with multiplicity k .
- If $P(x)$ is polynomial of degree n then it has at most n real roots/zeros (counting multiplicity).
- **Descartes's Rule of Signs:** If $P(x)$ is a polynomial written in descending order then,
 - The number of $P(x)$'s *positive* roots is equal to the number of $P(x)$'s sign changes or less than that by an even number.
 - The number of $P(x)$'s *negative* roots is equal to the number of $P(-x)$'s sign changes or less than that by an even number.

3. Let $P(x) = x^4 + 2x^3 - 5x^2 - 6x$.a. What is the largest possible number of roots for $P(x)$?b. What are the possible number of positive roots for $P(x)$?

- c. What are the possible number of negative roots for $P(x)$?